Simulation of macromolecular systems. Diffusion and Reaction processes.

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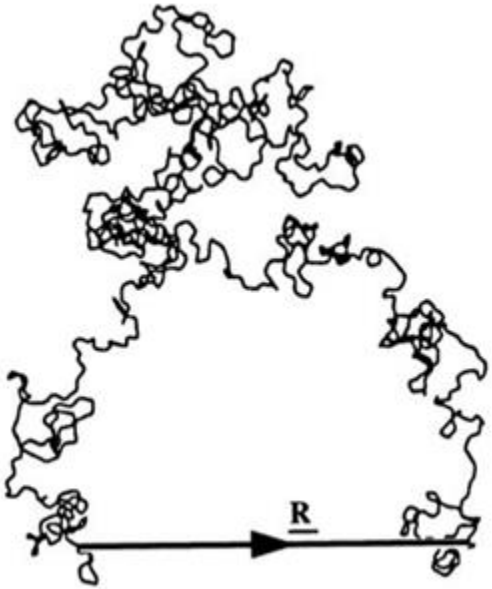
BioPhysChem group
www.ub.edu/biophyschem

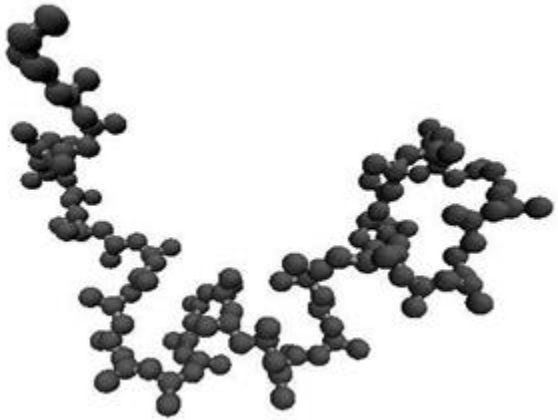
coarse-grained simulation models for polymers

Motivation: Structure, Properties?

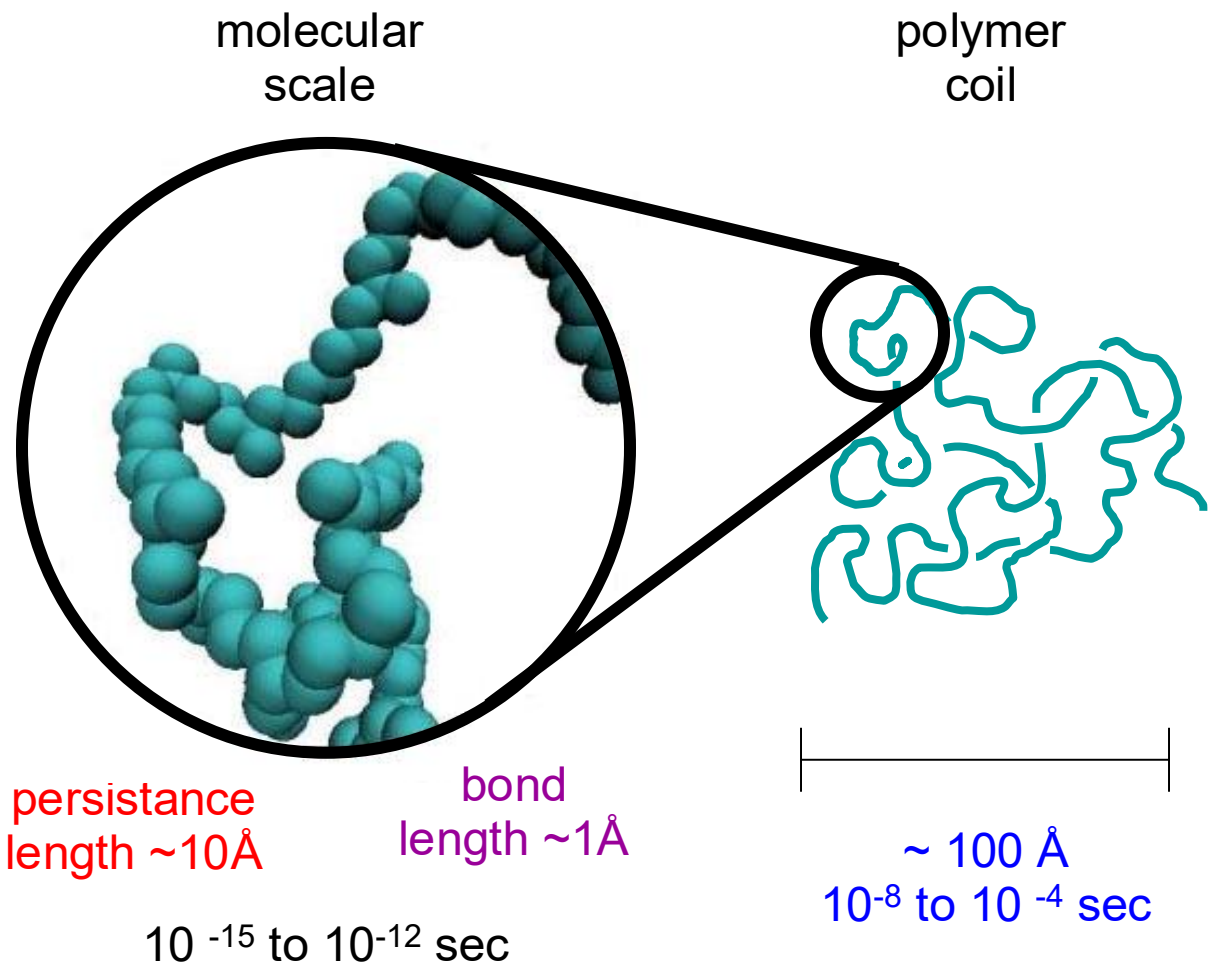
polyethylene



atactic-polypropylene

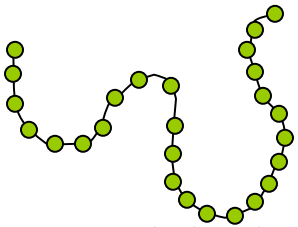


Hierarchy of Length and Time Scales

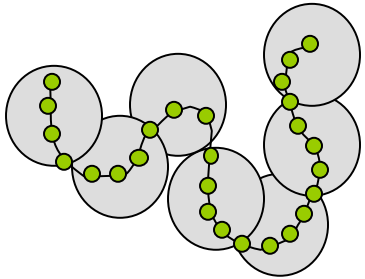


Coarse-graining of Polymer Simulations

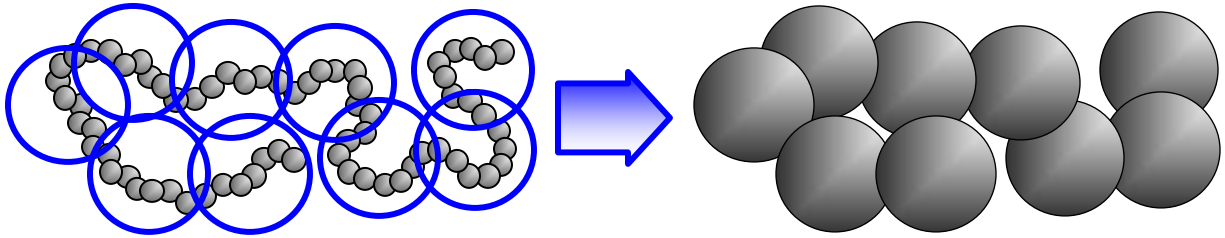
- Goal: To develop coarse-grained descriptions to access longer length and timescales



One CG particle describes *n* carbons of the detailed polymer

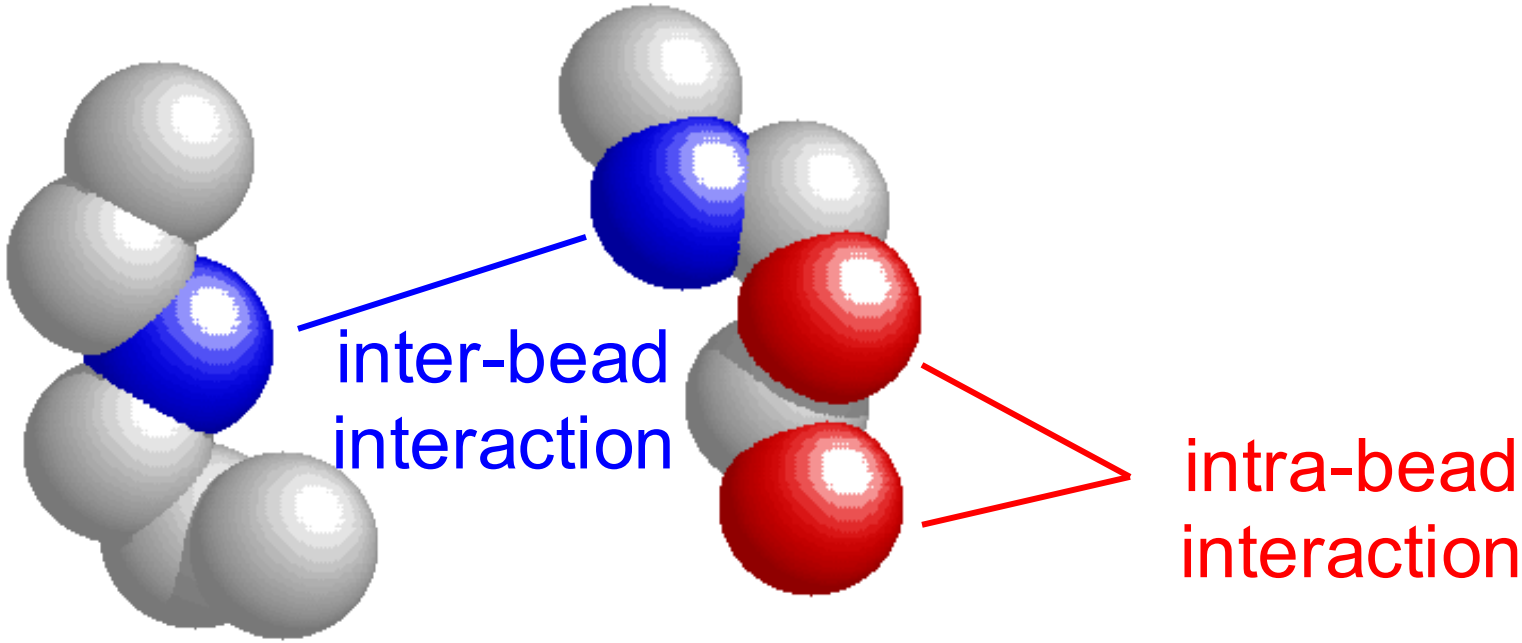


Coarse-graining method



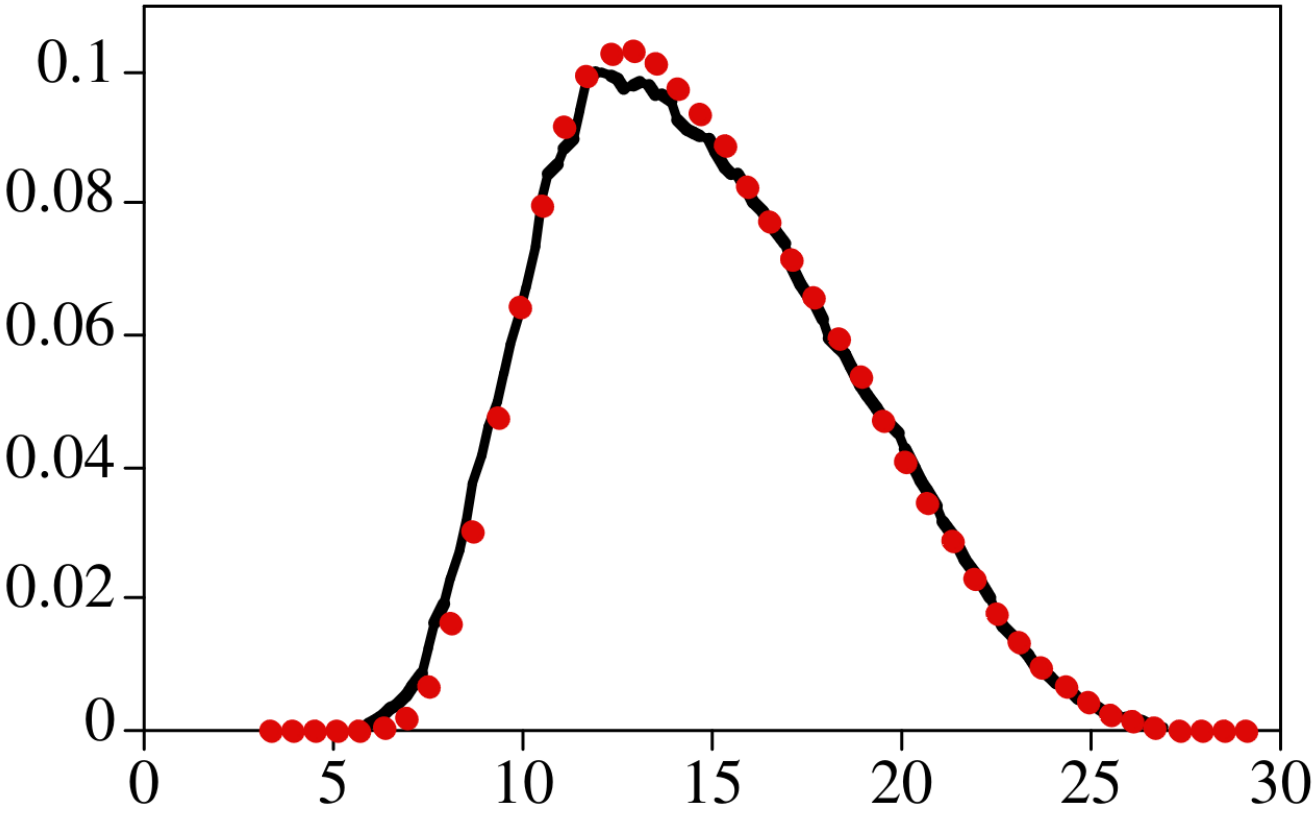
- Perform molecularly detailed simulations of polymers
- Define coarse-grained beads by grouping backbone monomers
- Calculate structural correlations between coarse-grained beads
- Determine effective bead-bead interactions that reproduce coarse-grained correlations of all-ato

Coarse Grained Potential



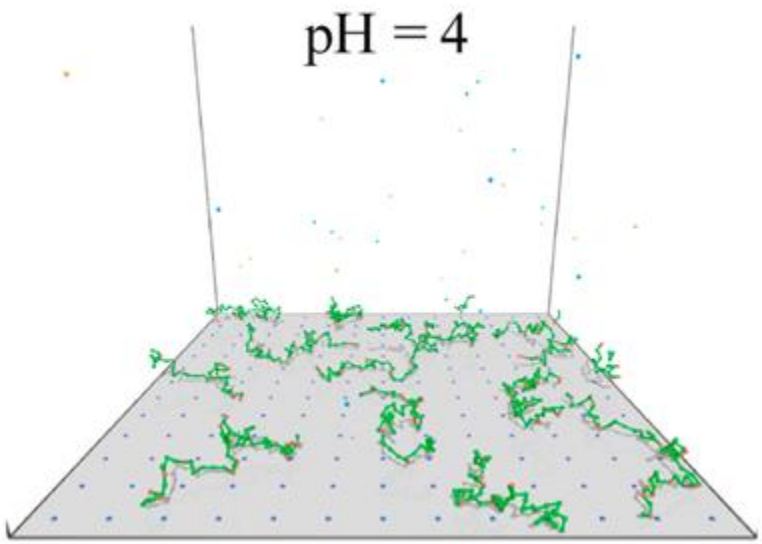
$$E_{total} = E_{inter} + E_{intra}$$

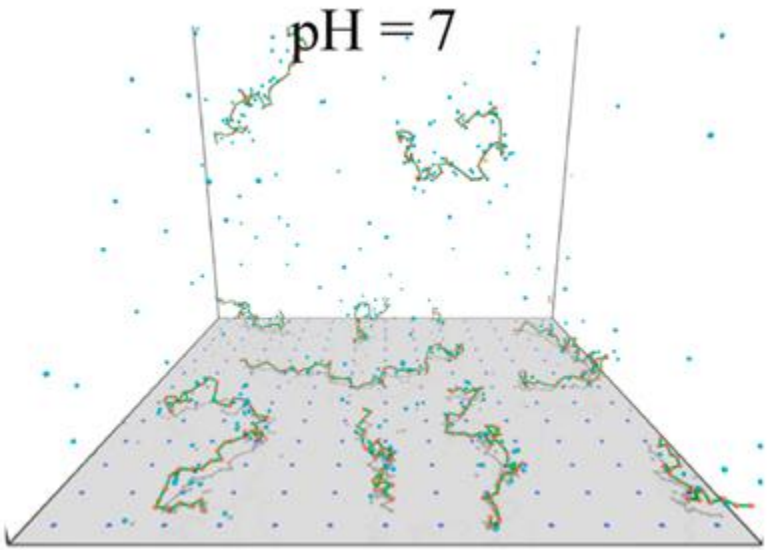
Oligomer conformation distribution



CG method reproduces conformational statistics of molecular oligomers

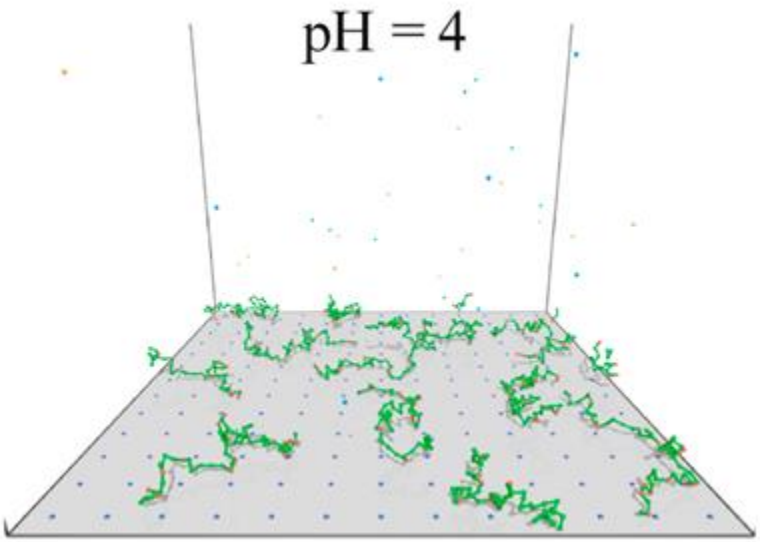
Weak-polyelectrolytes adsorption

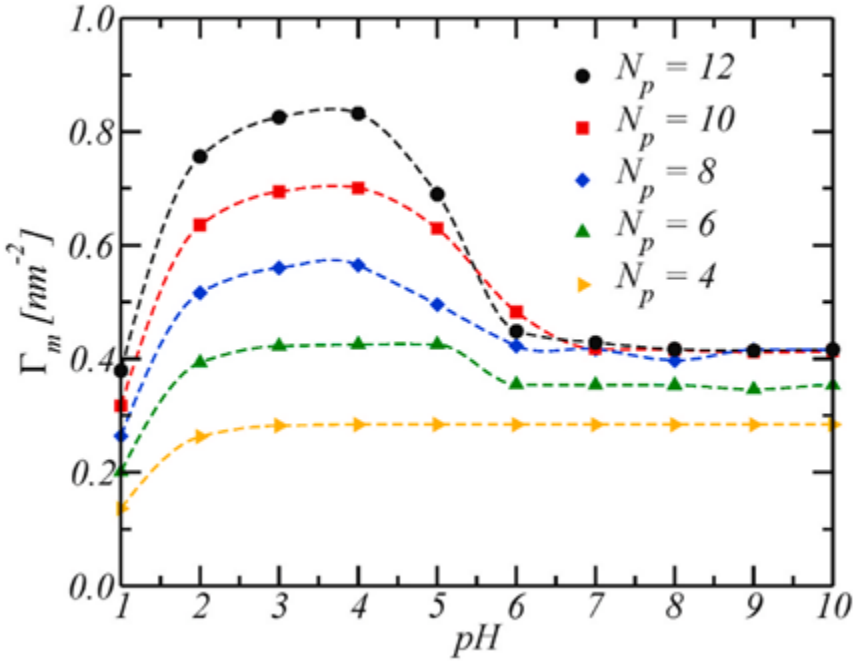




Narambuena, C. F., Blanco, P. M., Rodriguez, A., Rodriguez, D. E., Madurga, S., Garcés, J. L., & Mas, F. (2021). **Non-monotonic behavior of weak-polyelectrolytes adsorption on a cationic surface: A Monte Carlo simulation study.** *Polymer*, 212, 2020. https://doi.org/10.1016/j.polymer.2020.123170

Weak-polyelectrolytes adsorption





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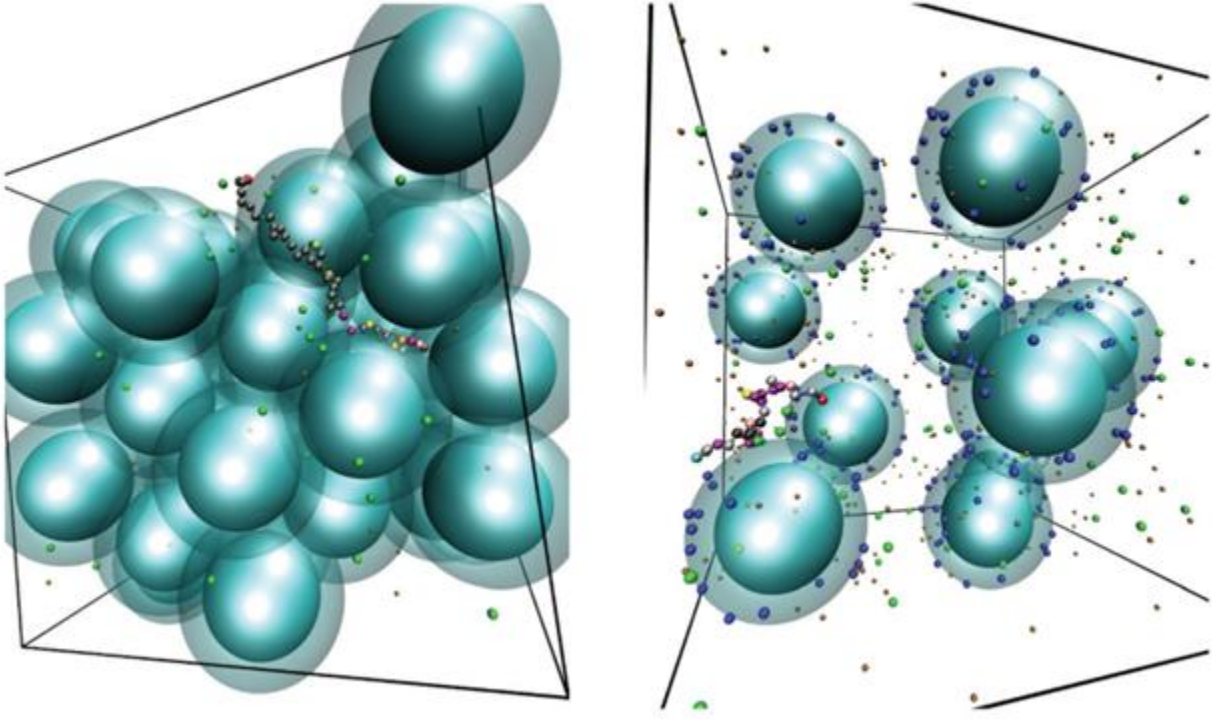
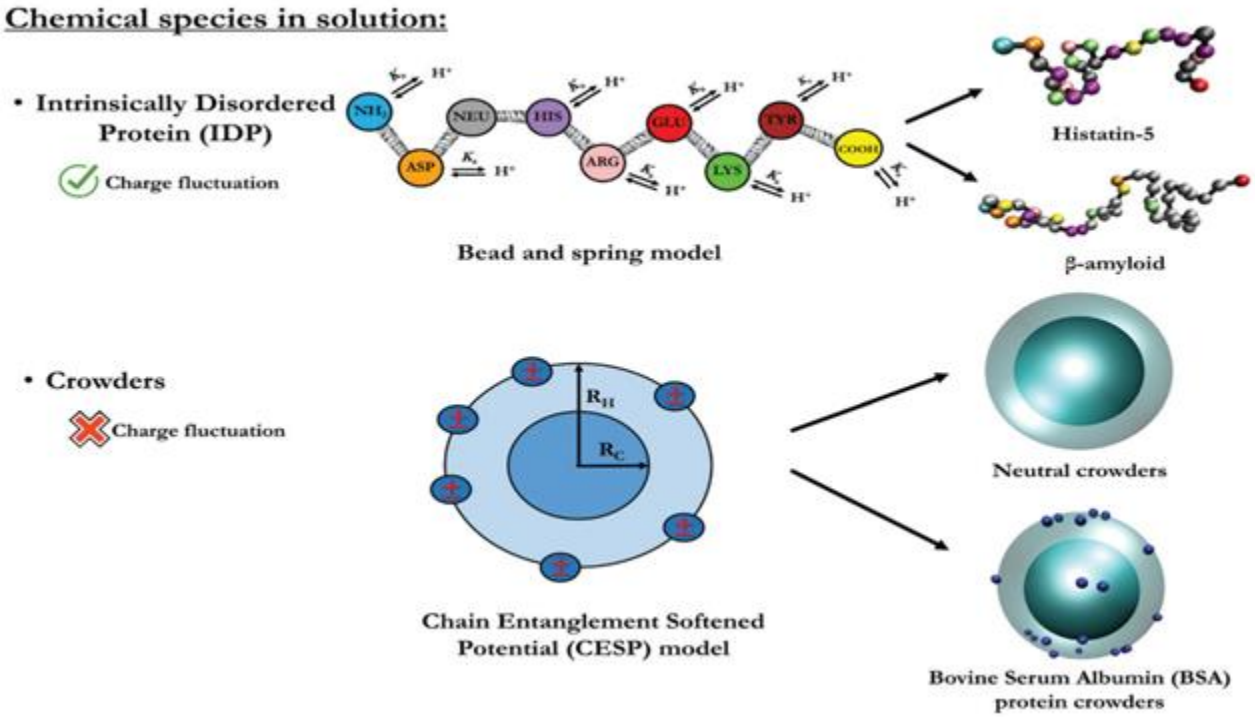
Macromolecular Crowding



Blanco, P. M., Madurga, S., Garcés, J. L., Mas, F., & Dias, R. S. **Influence of macromolecular crowding on the charge regulation of intrinsically disordered proteins.** *Soft Matter*, 655–669, 2021. https://doi.org/10.1039/d0sm01475c

Macromolecular Crowding



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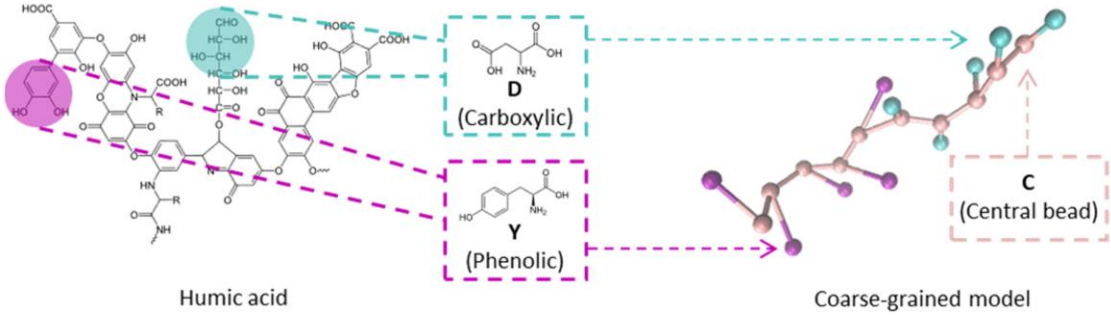
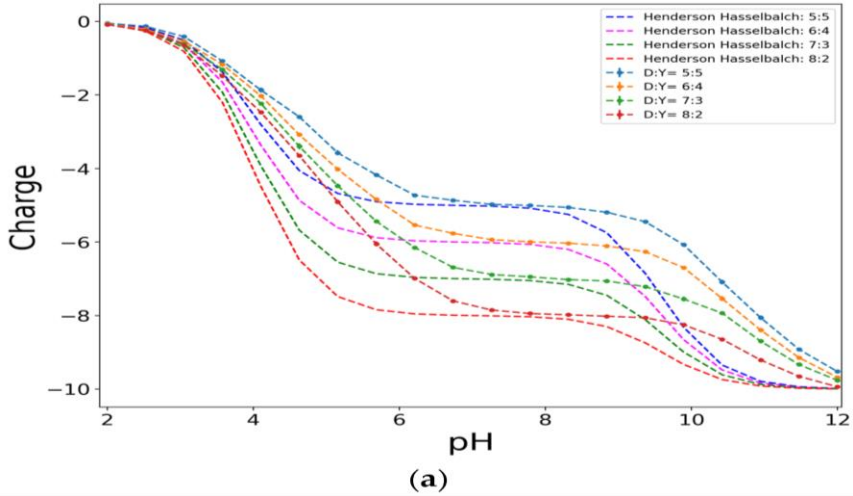
Constant-pH Simulations of a Coarse-Grained Model of Polyfunctional Weak Charged Biopolymers



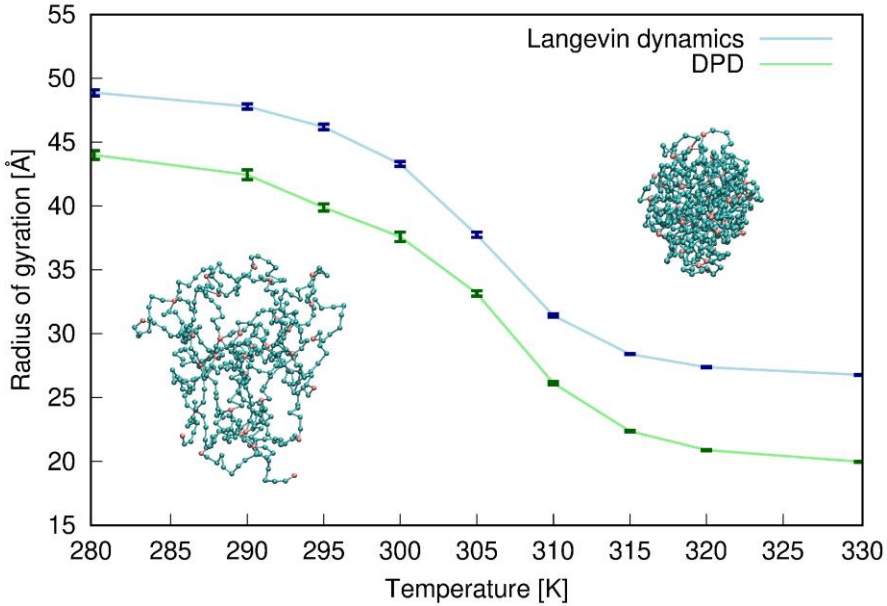
$$p_{acc} = min\left(1, exp\left(\frac{-\Delta U_{pot}}{k_B T}\xi(pH - pk_i)ln\,10\right)\right),$$



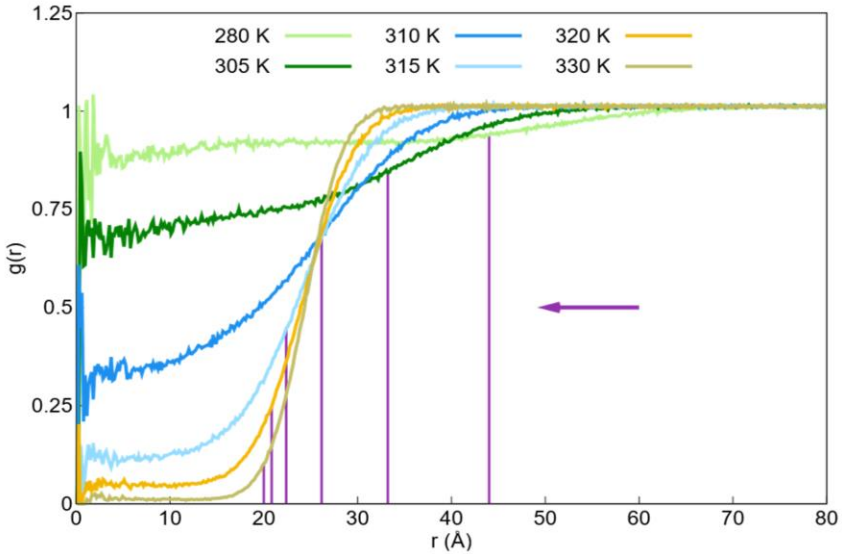
Effect of the D:Y proportion on the charge

Biophysica **2024**, *4*(1), 107-127;https://doi.org/10.3390/biophysica4010008

Temperature Dependence of a Thermosensitive Nanogel: A Dissipative Particle Dynamics Simulation of PNIPAM in Water



Radial distribution functions (g(r)) between the center of mass of the PNIPAM nanogel and water beads



Int. J. Mol. Sci.**2026**, *27*(3), 1241;https://doi.org/10.3390/ijms27031241